

INDIA IT SPENDING FORECAST (US\$ MILLION)

	2022 Spending	2022 Growth (%)	2023 Spending	2023 Growth (%)	2024 Spending	2024 Growth (%)
Data Centre Systems	3,607	18.6	3,766	4.4	4,121	9.4
Devices	49,853	2.0	46,309	-7.1	51,000	10.1
Software	13,115	15.1	14,931	13.8	17,689	18.5
IT Services	21,837	13.3	23,262	6.5	26,651	14.6
Communications Services	24,651	0.2	24,287	-1.5	25,148	3.5
Overall IT	113,063	5.5	112,554	-0.5	124,609	10.7

Source: Gartner (November 2023)

users of AI adopt a proper AI trust, risk, security management (TRISM) program at the company level. This involves integrating much-needed governance and proper guardrails upfront to proactively ensure that AI systems are compliant, fair, reliable, and protect data privacy. This is extremely important to avoid big mishaps in the future and will find a place in this year's IT agenda of companies using AI.

Demand for MLOps and ModelOps. With the increasing adoption of AI and machine learning (ML) by industries, the demand for machine learning operations (MLOps) services and professionals is also on the rise and is expected to increase further in 2024.

Traditional MLOps engineers might also have to skill up to learn the nuances of training genAI models as well, to manage the emerging paradigm of ModelOps! According to Gartner, ModelOps "is focused primarily on the governance and lifecycle management of a wide range of operationalised artificial intelligence and decision models, including machine learning, knowledge graphs, rules, optimisation, linguistic and agent-based models."

Service providers gear up to fill the gap. Forrester's Predictions 2024 report states that tech leaders will look to boost operations to grow with AI. According to the report, "Tech execs will seek

partners to close the AI gap as 50% fail to master it internally. Acquiring the knowledge and skill to deploy genAI, prompt engineering, scenario prioritisation, and new AI-infused applications will drive tech execs straight to their service providers."

Vendors such as Accenture, Tata Consultancy Services, and Wipro are already gearing up to bridge this gap. The report also suggests that tech executives will recognise this AI movement as an opportunity to rethink external relationships and will work to find a healthy balance between overly-dependent outsourcing and co-innovation partnerships.

Adaptive AI. While genAI is ideal for content creation, street-smart adaptive AI is best-suited for decision-making and process optimisation. It works beautifully in environments where change and complexity are constants, such as industrial monitoring, agricultural surveillance, fraud detection, investment and trading, healthcare, customer service, and robotics.

Hardware trends—A chip off the new block

Demand for generative AI chips.

In the Technology, Media, and Telecommunications (TMT) Predictions 2023 report, Deloitte predicts that the market for specialised chips optimised for generative AI

will exceed \$50 billion in 2024. The report suggests that generative AI chips will constitute two-thirds of the predicted global chip market of \$576 billion, with total AI chip sales reaching 11%.

Although generative AI chips are expected to become more accessible and affordable as supply increases, there are concerns about a potential bubble. Some fear that sales could be massive in 2023 and 2024, but actual enterprise generative AI use cases might not materialise, leading to a collapse in demand in 2025, similar to what happened with crypto mining chips in 2018 and 2021.

TrendForce also estimates that shipments of AI servers—including those equipped with graphics processing units (GPUs), field-programmable gate arrays (FPGAs), and application-specific integrated circuits (ASICs) will grow 38% in 2024. Apart from the major contenders like AMD and NVIDIA, we are now beginning to see major cloud service providers such as Google, AWS, Microsoft, and Meta taking giant strides towards developing their own proprietary AI chips. This might eat into the growth potential of GPUs.

General-purpose or customised, they are all going to need high bandwidth memory (HBM). In 2024, HBM3e, with speeds of up to 8Gbps, is likely to be a key enabler of AI compute cores. "With the growing complexity of training models and applications, demand for HBM is expected to skyrocket. It is projected that in 2024, HBM will make a significant contribution to memory suppliers' revenues given that the average unit price of HBM is several times higher than other DRAM products, for an annual growth rate of 172%" reads the TrendForce forecast.

Improved packaging technologies. After several years of research